Statement of Work

Introduction

The Statement of Work (SOW) document is used to clarify the project objectives, scope, tasks, timelines-deliverable, stakeholder management, and risk management of Active Vision project. The main purpose of this project is to provide valuable insights into the performance of tennis players and analyze in-game strategies. The key focus is on tracking the trajectory of tennis balls, which is critical for effective in-game analysis which could be used by amateurs who are interested in tennis and some senior professional tennis players. By accurately tracking the ball's movement, coaches can assess player performance, understand shot selection, and develop strategies to improve gameplay. The data collected can be used to identify patterns in player behavior, evaluate the effectiveness of different techniques, and provide feedback for training and match preparation. However, this task poses significant challenges due to the small size of the tennis ball and its high-speed movement, which can cause blurring or occlusion in some frames of the videos. Our model must learn and detect the ball's trajectory pattern accurately. Additionally, the project also evaluates players' performance by tracking their locations, poses, and swing angles during gameplay. Finally, cross-platform development may be carried out to make this analysis accessible and practical for users if time allows.

Project Objectives

The objective of the Active Vision project is to enhance the analysis of the performance of tennis players through advanced machine learning techniques. Specifically, the project aims to:

- **Develop Analytical Algorithms:** Creating robust algorithms to analyze captured motion, which tracks the high-speed and occluded trajectory of tennis balls accurately is required. Below are the main algorithms involved in this project, but they may be increased or reduced as the project progresses.

1. Playfield Detection Algorithm: It is mainly used to detect the boundary of the playfield so that the code can examine if the ball is out of bounds.

2. Human Extraction Algorithm: It helps the code to extract the position of players from the images or videos. The application can determine the location of the players for further analysis using the algorithm.

3. Human Key Points Algorithm: To denote the human's position, some key points are used to represent the whole human body. For example, the existing techniques typically use 16 points for the joints and head, to represent the key positions of the human skeleton.

4. Human Posture Estimation Algorithm: Based on the human key points algorithm, it is possible to utilize these key points to determine the posture of the player including forehand, backhand, serve and ready state.

5. Ball Tracking Algorithm: Enhance the real-time tracking of tennis ball trajectories, to accurately track the ball's movement and getting its location and speed during matches. We will fine-tune the promising existing ball tracking models such as TrackNet, combined with human posture estimation findings.

6. Court Detection Algorithm: Extract the keypoints from the court by fine-tuning promising pre-trained models, create and train with similar model architectures based on CNN is acceptable. Dataset of court should be carried out with data augmentation to prevent overfitting since all court from single angle looks the same.

- **Algorithm Optimization:** Since the application might be used in smartphones, specifically on devices supporting iOS and Android, it is necessary to optimize the algorithm to decreasing the size of the model to make the application runnable on the small devices. The goal is to ensure that the application can run smoothly on the following target devices while meeting the minimum viable product (MVP) performance requirements:

Target Devices:

- iPhone 14 and later models of iOS devices
- Android smartphones with Snapdragon 8 Gen 1 and above or MediaTek 9000 and above.
- PC or laptop attached with an RGB camera.

MVP Performance Requirements:

- Response Time: The algorithm's response time on target devices should be at least 30fps.
- Accuracy: The accuracy of ball trajectory tracking should be above 80%, and the accuracy of human posture recognition should be above 85%.
- Model Size: The optimized model size should be kept within 500MB to accommodate the storage and computational resources of mobile devices.

- **Application Development:** Application will be developed across multiple devices including iOS, Android and Web once the algorithm part has been completed. Flutter might be used for cross-platform development.

1. Real-Time Feedback: Implement a real-time feedback system to provide immediate insights and suggestions for technique improvement.
2. User-Friendly Interface: Develop an intuitive interface for users to interact with the system, view performance data, and track progress.

Project Scope

The Active Vision project aims to develop a state-of-the-art computer vision system designed to analyze tennis swing dynamics through comprehensive tracking of both the ball and the players. The project will involve three primary components:

1. Ball Tracking:

•Research: Research for existing models and how the ball tracking was being done. Promising models and datasets can be taken for fine-tuning to get the output we want (speed, location, state). The model can be directly used if the output fits our requirements.

•Data Collection: Collect open-source datasets related to tennis matches and other ball sports to prevent overfitting.

•Algorithm Development: Firstly, we need to build a model for tennis court detection. Based on the scope of the court detected, we can track the position of the ball much better. Then, design and implement algorithms for detecting and tracking the ball's position, speed, and trajectory.

•System Integration: Integrate the ball-tracking algorithms into the overall system and conduct system testing and optimization.

2. Player Tracking

•Object: The project aims to identify and track the player throughout the match, providing valuable insights into its movement, action, and trajectory.

•Methods:

i. Bounding Boxes: For detecting and localizing the object player within each video frame.

•Research Foundation: we conduct a thorough literature review to identify existing research papers, algorithms, and approaches that have been used for player detection, pose estimation, and action recognition in sports videos, particularly in tennis. Key studies such as Open Pose for real-time multi-person 2D pose estimation (Cao et al., 2018), Alpha Pose for accurate full-body pose estimation (Fang et al., 2016), and Dense Pose for dense human pose estimation (Guler, Neverova and Kokkinos, 2018) provide valuable insights into the detection and tracking of players' movements. For action recognition, methods like the Two-Stream Convolutional Networks (Simonyan and Zisserman, 2015), C3D (Tran et al., 2014), and Temporal Segment Networks (TSN) (Wang et al., 2016) offer robust frameworks for analyzing player actions, such as swings and running during a match.

This literature review will help in understanding the current state-of-the-art technologies and identifying gaps that the project can address.

•Data Collection: Collect open-source datasets related to human detection, tennis player's pose recognition and key points detection.

•Algorithm Development: Assess tennis players' performance by capturing and analyzing key metrics such as player locations, poses, and swing angles during gameplay.

•Outcome: By the later stage of the project, the goal is to have a working prototype that accurately tracks the players in sports videos, using the selected methodologies. By the detection result, we extracted the sports action, performance and some movement indicators to evaluate one player's performance which could be used in the interface to give users a good display.

3. Overall System Integration

•Integration: Combine the ball tracking and player tracking modules into a unified platform.

•User Interface Development: Ensure the trained networks and analytical tools are optimized for deployment on mobile devices, making the system accessible and user-friendly for coaches, players, and analysts.

•Comprehensive Testing: Conduct thorough testing in real-world environments to ensure the system meets functional and performance requirements.

Project Timeline

This project is roughly divided into two phases:

- Phase 1: Algorithm research (S2 2024)
- Phase 2: Application development (S1 2025)

This project will be divided into two sub-teams to maximize resource allocation and project efficiency:

- Ball tracking

- Players tracking

Both teams are responsible for gathering dataset, developing algorithms, and deploying models.

To guarantee that the development process is agile, project management tool Jira is introduced. A collective objective has been set and both teams will work on their respective components concurrently. The following timeline is for reference, actual development timeline will dynamically change based on client feedback and project progress.

Here is an estimated schedule from the model development to final publishing:

Table 1: Estimated schedule for Active Vision Project

Phase	**Task**	**Start Date**	**End Date**	**Duration (weeks)**	**Deliverable**
Literature review	1. Conduct a literature review on relevant state-of-the-art papers. 2. Select a suitable dataset for model training. 3. Draft a proposal on methodology of the project	29/07/24	12/08/24	2	Write a proposal on methodology
Data pre-processing, Model development	1. Pre-process dataset, training, valid, test set split and performing various data augmentation methods. 2. Develop the model either from scratch or by fine-tuning an existing public model.	13/08/24	23/09/24	6	Give precision, recall, and FM measurement scores on test set.
Result display	1. Display the in-game statistics of tennis and players on the test set video	24/09/24	15/10/24	3	Marking the statistics on video frames.

	frames. 2. Use the Qt package to display in-game statistics and player information on video frames.				
Final test and improvements. Results analysis and report writing	1. Perform edge case testing and evaluation, 2. Report our progress so far	16/10/24	30/10/24	2	Give the evaluation matric on how much the model has improved, draft the report
Quantize model	1. Quantize models to reduce model size and increase inference speed to fit mobile platform	23/02/25	09/03/25	2	Give the evaluation matric on precision, recall, Fm-score after quantization.
Model deployment	1. Read Qualcomm neural processing SDK and MediaTek NeuroPilot SDK manuals. 2. Deploy models on Android platform	10/03/25	21/04/25	6	Give the evaluation matric on precision, recall, Fm-score on Android platform
Software development	1. Develop software to perform real-time analytics of tennis matches. 2. The software should capture and analyze in-game matches from an Android smartphone's camera.	22/04/25	13/05/25	3	Show analytics on mobile devices.
Software testing	Perform unit testing	14/05/25	29/05/25	2	show unit testing results with 100% path and branch coverage on backend.

Milestones

Table 2: Project milestones across 2 semesters

Week	Description
s2 w3	1. Proposal of the methodology 2. dataset is prepared 3. Project Audit 1
s2 w6	1. Data pre-processing needs to be done. 2. The model begins to train. 3. Documents done for Audit 2
s2 w9	1. Fine-tune the model/hyperparameter 2. Report how much the algorithm has improved. 3. Get ready to show on Audit 3.
s2 w12	1. Complete model inference 2. Show output on test dataset.
s1 w3	1. Show results after the model is quantized
s1 w9	1. Complete model deployment 2. show a short demo of a real in-game tennis match.
s1 w12	1. Complete unit testing

Stakeholder Management Plan

Stakeholder Identify Result

Internal Stakeholders:

- Active Vision Project Team: Developers, designers, and project managers.
- Active Vision Technical Consultation Support Team: ANU TIME Lab, Arjun Raj, Francis Williams

External Stakeholders:

- Clients: Arjun Raj, Francis Williams.
- End Users: players, and analysts using the tennis analysis system.
- Project evaluators: Elena Williams (tutor), Charles Gretton (examiner)

- Project shadow team (KoPilot): Jiahao Xu, He Wang, Xuan Li, Xuke Du, Xiaojie Zhou, Jiawei Li, Xiangyu Bao

Stakeholder Analysis

Stakeholder Analysis is used to analyze all the individuals that are involved in Active Vision project who have a stake in the customer's strategy, and to help the customer identify the impact of major stakeholders on the strategy.

Stakeholder Mapping：

Power/Interest Grid: Categorize stakeholders based on their power (influence) and interest (impact on the project).

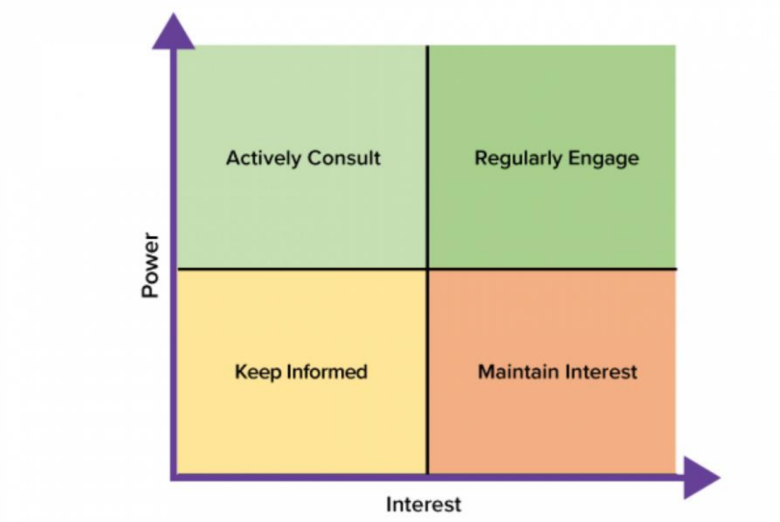

Fig 1: Stakeholders Power-Interest Matrix (Barton, 2021)

- **High Power, High Interest:** Clients, key project team members.
- **High Power, Low Interest:** Project sponsors, higher management.
- **Low Power, High Interest:** End users, support team, shadow team
- **Low Power, Low Interest:** General public, non-core team members.

Stakeholder’s needs and expectations

- **Clients:** Expect a robust system that provides detailed analytics to enhance player performance. The project needs to stay within the budget and be completed on schedule. Regular updates and demonstrations of progress are required.
- **End Users:** User-friendly interface, accurate tracking, and meaningful feedback. Desire reliability in the system’s performance data to make informed decisions about adjustments in technique.
- **Project Members:** Clear communication and documentation are required to align project objectives and progress. Besides, accessibility to advanced tools and resources helps to develop and test the system effectively.
- **Project Evaluators:** Tailored communication strategies that address the specific interests and needs of different stakeholder groups. Feedback and marks that are provided regularly allow the end users and clients to provide input on system functionality and usability, which can be used to improve the project continuously.
- **Support team members:** Require training and resources to troubleshoot and maintain the system efficiently. Clear protocols for escalating issues to development teams.
- **Shadow Team:** involved in asking questions along the product roadmap, validating our products and giving feedback. We shall be evaluating their performance as well, questioning any unclear statements and giving valuable feedback if required.

Communication Plan

To ensure effective collaboration and regular updates throughout the Active Vision project, some helpful communication methods will be implemented, including weekly updates to clients and the project team detailing progress and milestones.

- Emails/Discord: For formal updates and reports.
- Meetings: Hold regular weekly project status meetings with core team

and clients.

- Jira: Project Management Tools (for progress tracking and issue reporting)
- Workshops: For feedback and review sessions with end users.

Risk Management Plan for Active Vision

Introduction

This document outlines the Risk Management Plan for the Active Vision project. It includes procedures for identifying, assessing, responding to, and monitoring risks to ensure the successful completion of the project. This plan aims to address risks associated with these objectives to ensure the project's success and deliver a high-quality system.

The plan focuses on three key objectives:

1. Develop Analytical Algorithms: Create robust algorithms to analyze captured motion data, identifying key performance indicators (KPIs) such as swing speed, angle, and impact accuracy.
2. Real-Time Feedback: Implement a real-time feedback system to provide immediate insights and suggestions for technique improvement.
3. User-Friendly Interface: Develop an intuitive interface for users to interact with the system, view performance data, and track progress.

Risk Management Plan Progress

The Risk Management Plan will be updated regularly to reflect the status of risks and mitigation measures. Progress will be documented and reviewed to address any changes or emerging risks.

Risk Identification

- Definition

"According to the PMBOK Guide, risk identification is the process of determining what risks may affect the project and documenting their characteristics" (*PMBOK® Guide,* 2021).

- Methods for Risk Identification

Discussion: Regular team meetings and brainstorming sessions.

Historical Data: Learn from past historical projects to identify the same risks that may be encountered in current projects.
Expert Input: Consultations with project stakeholders and experts.

- Results

Details are captured in the Risk Register.

Risk Assessment

- Definition

Risk assessment involves evaluating the impact and likelihood of identified risks to understand their potential effect on the project.

- Tools/Standards

Risk Matrix**:** 'A risk matrix is a matrix that is used during risk assessment to define the level of risk by considering the category of likelihood against the category of consequence severity.' (Boogaard, 2022)

Likelihood-Qualitative Scale:

- Rare: May happen only in exceptional circumstances (less than 5% chance).
- Unlikely: Could happen but not expected (5%-20% chance).
- Possible: Might occur occasionally (20%-50% chance).
- Likely: Expected to occur frequently (50%-80% chance).
- Almost Certain: Expected to occur very frequently (more than 80% chance).

Harm severity-Qualitative Scale:

- Insignificant: Minor impact, no significant effect on project objectives.
- Minor: Noticeable impact but manageable without major changes.
- Moderate: Significant impact, requiring adjustments or additional resources.
- Major: Serious impact, affecting project outcomes and requiring substantial corrective actions.
- Catastrophic: Severe impact, potentially leading to project failure or significant harm.

Likelihood	Harm severity			
	Minor	Marginal	Critical	Catastrophic
Certain	High	High	Very high	Very high
Likely	Medium	High	High	Very high
Possible	Low	Medium	High	Very high
Unlikely	Low	Medium	Medium	High
Rare	Low	Low	Medium	Medium
Eliminated	Eliminated			

Fig 2: Risk Matrix Example (AuditBoard, n.d.)

Industry Standards: ISO 31000 or the PMBOK Guide for systematic risk assessment

- Results:

Details are captured in the Risk Register.

Risk Response Planning

- Definition

Risk response planning involves developing options and actions to enhance opportunities and reduce threats to project objectives.

- Methods

Mitigation Strategies: Develop specific actions to address high-impact risks.

Contingency Plans: Establish procedures for managing risks if they materialize.

- Results

Details are captured in the Risk Register.

Risk Monitoring/Risk Control

- Definition

Risk monitoring involves tracking identified risks, monitoring residual risks, and identifying new risks.

- Methods

Risk Monitoring: Regular reviews and updates of the Risk Register.

Communication: Frequent updates and consultations with stakeholders.

Control Measures: Implement preventive actions and continuously educate relevant parties.

- Results

Details are captured in the Risk Register.

Risk Register

The Risk Register will be the central repository for all risk-related information, providing a comprehensive overview of identified risks, their assessments, responses, and monitoring activities.

Table 3: Risk estimation and solution for Active Vision project

Risk	Risk Information	Impact	Risk Assessment (Likelihood)	Risk Assessment **(Harm severity)**	Preventive Measures	Risk treatment	Principle
Personal change	Lack of manpower affects delivery	Huge	Likely	Major	Set aside more time	Postpone the project	Project leader

Requirement Changes	Changes or ambiguity in client requirements affecting project goals	Huge	Likely	Major	Communicate with the clients regularly	Reassign the tasks	Project leader
Model Performance Issues	Model accuracy or performance below expectations in actual application	Huge	Likely	Major	Evaluate the models before we decide to use	Adjust the weights or change models	Project leader
Client Satisfaction	Client dissatisfaction affecting long-term success and relationships	Huge	Likely	Major	Always take positive actions to clients' feedback	Communicate with the clients friendly	Project leader
Data Set Quality Issues	Poor quality of data used for training and testing	Huge	Likely	Major	Evaluate the dataset before use	Change to another datasets	Dataset collectors
Algorithm Incompatibility	Existing algorithms may not meet project needs or be incompatible with target hardware	Huge	Possible	Moderate	Infer and demonstrate the algorithms manually before use	Come up with another algorithm	All members
Integration Issues	Problems integrating model with other system components	Huge	Possible	Moderate	Evaluate the compatibility of models before use	Change to other models	Model proposer
Computational Resource Limitations	Insufficient computational resources affecting model training and inference speed	Medium	Possible	Moderate	Consider using the lightweight models	Seek for high-performing GPUs and CPUs	Project leader

Client Changes	Changes in client demand or client number affect resource allocation	Medium	**Unlikely**	Moderate	Communicate with the clients regularly	Negotiate with the clients to reduce the negative impacts	All members
Equipment Failure	Failure or damage to required hardware or equipment	Medium	Possible	Moderate	Evaluate the equipment before use	Change to other equipment	Project leader

Resources

- Due to the complexity and challenges involved in developing and integrating models for this project, additional guidance and technical consultation will be necessary. The members of the ANU TIME lab would be our valuable resource to provide technical consultation.
- We would use the open-source dataset to train and evaluate our models. The dataset we plan to use has been documented here.
- We would comply with the License of development tools such as PyCharm, Visual Studio Code and any other software.

Agreement

Client section:

Name	Signature	Date (DD/MM/YYYY)
Arjun Raj	Arjun Raj	06/08/2024
Francis Williams	*Francis Williams*	06/08/2024

Group Members section:

Name	Signature	Date (DD/MM/YYYY)
Tao Lu	Tao Lu	08/04/24
Xi Ding	Xi Ding	08/04/24
YiChi Zhang	Yichi Zhang	08/04/24
Xing Chen Zhang	ZXC	08/04/24
Zhiyuan Lu	陆之远	08/04/24
Linxiang Hu	胡霖翔	08/04/24
Pei Ling Lam	Peiling	08/04/24

References

Barton, S. (2022, January 18). *Stakeholder X personas*. The Nerdy BA, [Image]. https://www.thenerdyba.com/post/stakeholder-x-personas

Boogaard, K. (2022, May 18). *What is a risk matrix? (With example) | Wrike*. Blog Wrike. https://www.wrike.com/blog/what-is-risk-matrix/

PMBOK® Guide. (2021). Project Management Institute. https://www.pmi.org/pmbok-guide-standards/foundational/pmbok

The risk assessment matrix: What is it and why is it important? (n.d.). Audit, Compliance, & Risk Management Software | AuditBoard [Image]. https://www.auditboard.com/blog/what-is-a-risk-assessment-matrix/

Cao, Z., Simon, T., Wei, S. and Sheikh, Y., 2018. OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields. arXiv preprint arXiv:1812.08008. Available at: https://arxiv.org/abs/1812.08008 [Accessed 3 Aug. 2024].

Fang, H., Xie, S., Tai, Y.-W. and Lu, C., 2016. AlphaPose: Real-Time and Accurate Full-Body Multi-Person Pose Estimation and Tracking in the Wild. arXiv preprint arXiv:1612.00137. Available at: https://arxiv.org/abs/1612.00137 [Accessed 3 Aug. 2024].

Guler, R.A., Neverova, N. and Kokkinos, I., 2018. DensePose: Dense Human Pose Estimation In The Wild. arXiv preprint arXiv:1802.00434. Available at: https://arxiv.org/abs/1802.00434 [Accessed 3 Aug. 2024].

Tran, D., Bourdev, L., Fergus, R., Torresani, L. and Paluri, M., 2014. C3D: Generic Features for Video Analysis. arXiv preprint arXiv:1412.0767. Available at: https://arxiv.org/abs/1412.0767 [Accessed 3 Aug. 2024].

Wang, L., Xiong, Y., Wang, Z., Qiao, Y., Lin, D., Tang, X. and Van Gool, L., 2016. TSN: Temporal Segment Networks for Action Recognition in Videos. arXiv preprint arXiv:1608.00859. Available at: https://arxiv.org/abs/1608.00859 [Accessed 3 Aug. 2024].